

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Pipeline Blowdown and Depressurisation Plan

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Revision History

Rev No.	Date	Description of Revision
R01	28.07.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-PLN-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001- 3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
API	American Petroleum Institute
BFPCL	Brass Fertilizer and Petrochemical Company Limited
BOD	Basis of Design
DEP	Design and Engineering Practice
DPR (now NUPRC)	Department for Petroleum Resources
FEED	Front End Engineering Design
GPP	Gas Processing Plant
MMSCFD	Metric Million Standard Cubic Feet Per Day
MTPD	Metric Tonnes Per Day
NUPRC (formerly DPR)	Nigerian Upstream Petroleum Regulatory Commission
OML	Oil Mining License
PIG	Pipeline Inspection Gauge
PMC	Project Management Consultant
RTU	Remote Telemetry Unit
SPDC	Shell Petroleum Development Company

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

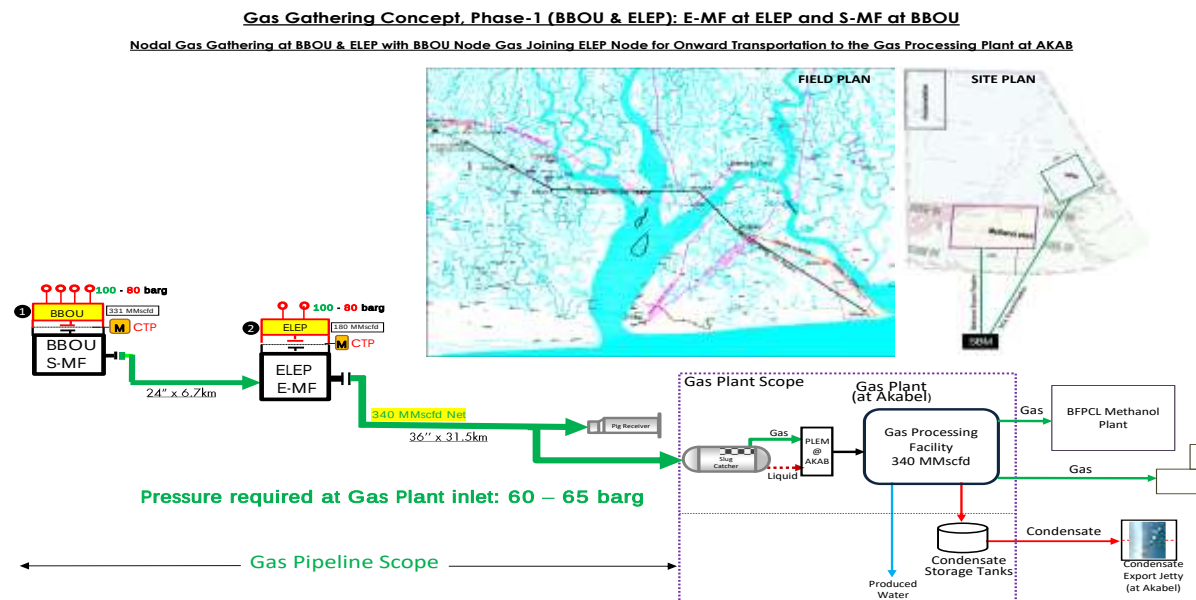


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to present the plans for the relief, vent, blowdown, and flare systems for the FEED for Gas Gathering Pipelines & Associated Infrastructure Phase 1 Project.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Relief System Design Philosophy

3.1. General

1. Basic concepts of the safety system, protection methods, and requirements of the Facility and systems shall be in accordance with API 14C.
2. When equipment or systems can be exposed to pressure that would exceed the design pressure of that equipment or system, protection shall be provided by relief devices and in accordance with the document EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-002_Process Control & Safeguarding Philosophy.
3. Vapor depressuring (blowdown) devices shall be used to reduce potential for stress rupture failure during a fire and to mitigate the consequences of a vessel or pipe leak by reducing inventory and leakage rate.
4. Selection and sizing of relief devices shall be in accordance with API 520 and API 526.

3.2. Causes / Scenarios

Following cases or scenarios are to be considered alongside Manifold pressure relief cases.

- Blowdown of inventory between the inlet SDVs and outlet SDV at each Manifold location;
- Manual depressurization of inventory inlet SDVs and outlet SDV at each Manifold location;
- Manual depressurization of the 24" pipeline from BBOU to Elepa.
- Manual depressurization of the 36" pipeline from Elepa to Akabel.

3.3. Overpressure Protection

1. Possible causes of overpressure and required relief capacity shall be determined in accordance with the document EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-002_Process Control & Safeguarding Philosophy.
 - a. For atmospheric and low pressure storage tanks, determination of venting requirements and pressure protection shall be in accordance with API 2000. This will apply to odorizing tank and condensate tank.
 - b. Good engineering judgment, as discussed in ASME BPVC Section VIII Division 1, "Forward" section, shall be exercised to address other overpressure causes that may reasonably arise from operation of the facility.
2. Each credible relief contingency shall be considered and the required relieving capacity shall be determined and documented for each protected system.
3. Per API 521, "Relief Load Causes," simultaneous occurrence of two or more conditions that

could result in overpressure shall not be considered if the causes are unrelated.

- a. A single event, such as an external fire that could affect multiple vessels, shall be analyzed as a common-mode, multiple-vessel event when sizing flares and flare headers.
 - b. Pressure relief devices and associated piping and headers shall be designed for the worst-case event or events.
4. Equipment connected to other equipment without isolation may be protected by a common relief device; however, each piece of equipment shall be included in the analysis when sizing the common relief system.

3.4. Selection of Relief Devices

1. Selection and sizing of relief devices shall be in accordance with API 520, API 2000 and API 526.
2. Some pressure relief valves contain components that are not full flange rated. Pressure relief valve discharge piping and flare system design shall ensure the following:
 - a. Maximum backpressure shall not exceed the mechanical integrity limits of pressure relief valves, as defined in API 526.
 - b. Maximum backpressure shall not impair pressure relief valve capacity required to protect upstream equipment and piping systems.

3.5. Pressure Relief Valve Spare, Piping, and Isolation Requirements

Pressure relief valve inlet and discharge piping and isolation requirements shall be in accordance with API 520 and API 521.

3.5.1. Spares

Spare pressure relief valves shall be installed in accordance with criteria EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-002_Process Control & Safeguarding Philosophy.

3.5.2. Inlet Piping

Pressure relief valve inlet piping shall be designed in accordance with API 521.

3.5.3. Discharge Piping

1. Discharge piping from pressure relief devices shall be designed in accordance with API 521.
2. The maximum velocity in gas discharge piping should not exceed a Mach number of 0.9 except at a tailpipe exit to the atmosphere. If velocity in gas discharge piping could exceed a Mach number of 0.9, then acoustically induced vibration (AIV) analysis of the discharge

piping shall be undertaken and TGNL authorization shall be obtained.

3. Discharge piping shall be self-draining toward a designated low point and shall have a minimum slope of 1:200.
4. Weep holes or liquid collection systems shall be provided on vents to atmosphere.
5. The Joule-Thomson effect occurring across a pressure relief valve when relieving can lower the temperature so that it is within the hydrate formation region.
 - a. Due to the high velocity when relieving, blockage is not normally a concern.
 - b. However, valve blockage due to small leaks can occur.
 - c. Heat tracing or hydrate inhibition shall be provided if blockage is a concern.
6. Low temperature evaluation of the pressure relief valve and discharge piping shall be performed to prevent brittle fracture.

4. Vent Systems

4.1. General

1. Flare and vent systems shall be designed in accordance with API 521 recommendations.
2. Continuous venting shall be minimized.

4.2. Vents to Atmosphere

1. All hydrocarbon gas shall be contained such that venting is occasionally carried out as a result of pressure relief operations or blowdown in emergency situation only.
2. All relief valves and blowdown valves shall be channelled to a common relief header that terminates at a vent.
3. Vents from potentially hydrocarbon-bearing systems shall be combined and routed to a safe location for proper disposal. This location shall be at least 10 feet (3 m) from electrically hazardous classified areas and at least 25 feet (7.6 m) from air intakes (engine or heating, ventilation, and air conditioning [HVAC]).
4. Requirement for a flame arrestor or detonation arrestor shall be considered for each vent to atmosphere application.
5. Vents from non-hazardous utility systems may be disposed of individually to a safe area near the source.
6. Vents shall be evaluated for dispersion of combustible gases in the unignited condition. This minimizes the risk that combustible gas mixtures could occur at locations considered as safe areas at the Facility or where human activities such as smoking and burning could occur.
7. Location and orientation of the atmospheric discharge shall be evaluated using appropriate vapor dispersion models and thermal radiation models. A common vent system may be provided if the magnitude of the release is significant or if liquid knockout and retention is required.
8. Dispersion modeling shall consider design flow and low flow conditions from vent sources. Model shall consider prevailing wind under stable atmospheric conditions.

4.3. Vent Load Analysis and Sizing Criteria

1. Vent header shall be sized for the maximum expected load. Maximum load is the flow rate that imposes the highest pressure drop in the system.
 - Lines from individual relief device shall be sized for the rated flow from that device.
 - A main header or sub-headers shall be sized for a specific maximum contingency.

2. For maximum load analysis, careful consideration shall be given to potential process upsets or emergency conditions that could affect several vessels or systems and cause them to relieve simultaneously. Normally, the following contingencies shall be included in the analysis:
 - Relief rates from a total utility failure such as instrument air, cooling water, or electrical power.
 - Relief rates from a partial utility failure.
 - Maximum relief rate from a single item failure.
 - Maximum rate during emergency blowdown.
 - Maximum rate during an emergency blowdown that can also cause the blowdown of rotating equipment (e.g., centrifugal compressor).
 - Maximum rate in the event of fire.
 - Time limits on controlled depressuring of flowlines and pipelines.
3. Vent headers are normally sized based on backpressure limitations of the relief devices, and gas velocity in headers should not exceed a Mach number of 0.9.

4.4. Closed Drain Tank and Liquid Disposal

1. Vent knockout drums shall be sized based on the following:
 - a. Separate the liquid from the gas before it is relieved to atmosphere through a vent stack per API 521.
 - b. Hold the maximum amount of liquid discharge during an emergency for 20–30 minutes per API 521.
 - c. If the knockout drum also functions as a closed drain vessel, the pump out capacity shall exceed the maximum possible drainage rate of any vessel when drained at atmospheric pressure.
1. The maximum gas relief case may not coincide with the maximum liquid relief case. Knockout drum shall be sized for the maximum gas relief case and the maximum liquid relief case.
2. Vent stack knockout drums shall be sized for maximum droplet diameter at the outlet of 300 μm .
3. The selection of a horizontal or vertical knockout drum shall be based on economic consideration, considering the required slope of the header and the maximum liquid that must be contained.
4. The knockout drum shall be designed as a pressure vessel with minimum design pressure of 100 psig (690 kPa gauge).

4.5. Vent Structure

1. The size and height of a vent stack shall be determined based on the following operational and environmental factors:
 - a. Acceptable dispersion levels
 - b. Acceptable noise levels
 - c. Velocity and backpressure limits

4.5.1. Dispersion

1. Atmospheric hydrocarbon releases shall be at a safe location so that personnel are not exposed to a hazardous condition and facilities are not exposed to unexpected corrosion or damage.
2. In general, atmospheric release shall meet the following criteria:
 - a. Concentration of flammable components is less than 50 percent of LFL before it reaches any potential ignition source. Depending on the averaging time selected when performing dispersion calculations, the calculated distribution of a vapor cloud to its LFL will vary.
 - b. No noticeable stench or irritation levels, both inside and outside the facility property limit.
3. Dispersion calculation shall be performed for vent stack, and significant hydrocarbon vents, to confirm the flammable gas concentrations and hazardous gas levels are below acceptable limits.

4.5.2. Noise Limits

Noise level shall not exceed 60 dB limit at distance of 10m from the Vent opening.